INCH POUND

MIL-C-3098/87D
5 May 1993
SUPERSEDING
MIL-C-3098/87C
12 September 1985

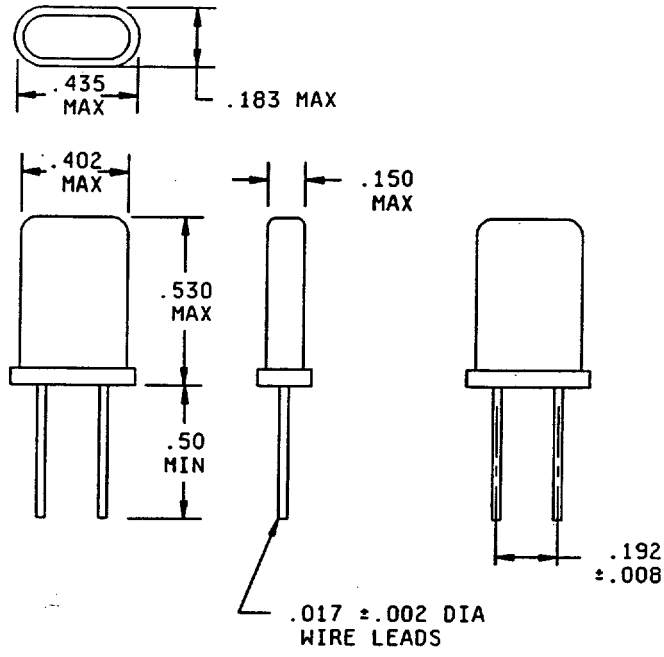
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR111/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 17 to 62 MHz; third mechanical overtone; noncontrolled; series resonance.



Inches	mm
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR111/U.

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REQUIREMENTS:

Holder: HC-18/U (MIL-H-10056/9) or HC-43/U.

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17 to 62 MHz, inclusive.

Capacitance, shunt: 4 pF, maximum.

Mode of oscillation: Third mechanical overtone.

Rated drive level: 2.0 ±0.4 mW.

Reference temperature: +30°C ±1°C.

Operating temperature range: -55°C to +105°C, inclusive.

Frequency tolerances:

Operating temperature range: ±50 ppm.

Reference temperature: ±15 ppm.

Equivalent resistance: 40 ohms, maximum.

Unwanted modes: Method II, the minimum unwanted mode effective resistance shall be at least two times that of the main mode, or 40 ohms, whichever is greater.

Shock (specified pulse): MIL-STD-202, method 213, with the following exceptions, 18 impact shocks of 5 g's each, time duration 11 ±1 milliseconds. Three shocks shall be applied in each direction along each of three mutually perpendicular planes.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent, or 2 ohms, whichever is greater.

Vibration: MIL-STD-202, method 204, test condition A, 0.03 inch double amplitude or 5 g's, whichever is less.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent, or 2 ohms, whichever is greater.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent, or 2 ohms, whichever is greater.

Aging:

Frequency change permitted: ±5 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
DLA - ES

User activities:
Air Force - 17

Preparing activity:
Army - ER

Agent:
DLA - ES

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